

Groundedness Is Not Truth: Evidence-Chain Forensics for Retrieval-Augmented Generation

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Abstract

A single faithfulness score can hide failures that require different repairs: a RAG answer may be grounded in text whose source is stale, incomplete, contradicted, low-authority, or insufficient. We introduce **CONTEXTTRACE**, a deterministic, local-first SDK and CLI for evidence-chain forensics that separates claim support, citation state, source condition, abstention need, root cause, repair, and regression prevention. We also introduce **Groundedness-Truth-Gap**, a suite for controlled conformance, while frozen RAGTruth and stress tests serve as failure-safety controls. ContextTrace’s central contribution is an inspectable evidence-chain contract. On a fresh 84-case public-document evaluation, it catches 76 of 77 failure-bearing cases and preserves all seven clean cases (balanced accuracy 0.994; MCC 0.929), the same binary confusion counts as a frozen GPT-4.1-mini judge, while producing typed diagnostic records without additional model calls. The clean-case sample is small, and fine-grained label, root, and unverifiable attribution remains limited. Post-review counterfactuals further show strong dependence on source metadata and temporal cues. We release a blinded 50-case annotation packet and scorer; independent adjudication remains pending. Our claim is reproducible fail-closed diagnosis, not truth verification or broad evaluator superiority.

1 Introduction

Retrieval-augmented generation (RAG) conditions generation on external documents (Lewis et al., 2020), but grounding in retrieved text is not the same as truth. An answer can quote its source exactly and still be unsafe when that source is stale, incomplete, contradicted by a stronger source, or silent on the requested fact. Scalar faithfulness does not identify these distinctions or the component that should change.

We study *evidence-chain forensics*: reconstructing a verifiable path from each answer claim

through retrieved evidence and citation edges to an observable failure and a concrete repair. This framing requires more than detecting hallucination. It requires preserving intermediate decisions, refusing to impute diagnostics a baseline does not expose, and distinguishing grounding in supplied evidence from real-world truth.

CONTEXTTRACE implements this framing as a deterministic, local-first pipeline. It consumes portable RAG or agent traces and emits claim verdicts, minimal evidence spans, citation states, source assessments, abstention decisions, failure labels, primary root causes, and evidence-backed repair and regression suggestions. Figure 1 summarizes the pipeline.

Our contributions are threefold. First, we formalize evidence-chain forensics for RAG, separating groundedness, source condition, citation validity, abstention need, root cause, and regression prevention. Second, we implement ContextTrace as a local, deterministic SDK/CLI that converts RAG traces into inspectable claim-to-evidence-to-repair records. Third, we introduce Groundedness-Truth-Gap and evaluate the system with fresh public-document traces, frozen RAGTruth controls, adversarial stress tests, and same-ID baselines. The resulting evidence supports fail-closed diagnosis and operational traceability, while fine-grained naturalistic attribution remains limited. We release a blinded 50-case annotation packet and scoring protocol spanning naturalistic, focused-benchmark, and RAGTruth-derived cases.

2 Related Work

RAG evaluation. RAGAS decomposes automatic evaluation into faithfulness, answer relevance, and context relevance (Es et al., 2024). ARES uses synthetic training data and lightweight judges for context relevance, answer faithfulness, and answer relevance (Saad-Falcon et al., 2024).

DeepEval provides a software-oriented collection of LLM evaluation metrics (Confident AI, 2026). These tools make automatic testing practical, but their outputs are not interchangeable with root-cause, citation, and span diagnostics. Our baseline tables therefore score only common outputs and mark unsupported fields N/A.

Diagnostic benchmarks. RAGChecker introduces fine-grained retriever and generator diagnostics (Ru et al., 2024), the closest existing analogue to our contract, but its diagnostics characterize retrieval and generation quality and do not model source condition – staleness, authority, or canonical-source status – so a RAGChecker-clean case can still cite a source that is stale or superseded. RAGTruth supplies word-level hallucination annotations across RAG tasks (Niu et al., 2024), while CRAG evaluates factual and temporally dynamic retrieval settings (Yang et al., 2024). ContextTrace uses RAGTruth as external calibration evidence and CRAG as a secondary track; it adds an explicit serialized chain from claims to repairs.

Recent systems overlap more directly with actionable diagnosis. RAGXplain uses LLM reasoning to translate metric assessments into explanations and repair recommendations (Cohen et al., 2025). RAGTrace is an interactive system for exploring retrieval–generation dynamics and locating failures (Cheng et al., 2025). A 2026 enterprise benchmark proposes a multi-dimensional taxonomy spanning reasoning, retrieval difficulty, source structure, and operational explainability (Narita et al., 2026). ContextTrace differs by emitting deterministic, replayable, local claim–evidence–source-condition records and CI regression targets, rather than LLM-generated recommendations or primarily interactive exploration. These are complementary system designs; our experiments do not establish superior human actionability.

Model judges and observability. LLM judges can approximate human preferences (Zheng et al., 2023), but order, self-enhancement, and fairness biases remain material (Wang et al., 2024). We therefore report our model-based review solely as a simulated pilot. At the systems layer, OpenTelemetry defines vendor-neutral trace concepts (Cloud Native Computing Foundation, 2026), and OpenInference extends semantic conventions to AI workloads (Arize AI, 2026). ContextTrace accepts such trace-like records but targets semantic evidence diagnosis rather than telemetry trans-

port. TRAIL further shows that issue localization in agent workflows is a distinct and difficult task, providing 148 human-annotated OpenTelemetry-structured traces and a fine-grained error taxonomy (Deshpande et al., 2025). We treat TRAIL as the appropriate external transfer benchmark for future agent-trace claims; the present evaluation is predominantly RAG and does not establish agent-trace generalization.

3 Evidence-Chain Forensics

Let a trace be $T = (q, a, C, Z, M)$, where q is a query, a an answer, $C = \{c_j\}$ retrieved contexts, Z citation edges, and M optional source and execution metadata. Claim decomposition produces $A = \{a_i\}$. For each claim, the verifier emits

$$y_i = (v_i, e_i, z_i, s_i, r_i, f_i), \quad (1)$$

where v_i is a support verdict, e_i a localized evidence span, z_i a citation state, s_i a source-state assessment, r_i a reason, and f_i a claim-level fix. An aggregate diagnosis $D(T)$ contains failure labels, one primary observable root cause, an abstention decision, and a regression target.

Evidence-chain forensics imposes three invariants. *Traceability*: a supported verdict points to source text. *Coverage honesty*: unavailable outputs remain N/A rather than zero or inferred. *Repair locality*: root causes describe the earliest observable component supported by the trace, not hidden implementation state. For example, missing evidence can support “retrieval miss” only when a supplied corpus or retrieval contract establishes that the evidence should have been available; otherwise the safer label is “should have abstained” or “corpus gap.”

Figure 2 illustrates the central distinction. The answer is verbatim-supported by an archived policy, but a current canonical policy gives a different value. The correct diagnosis must preserve both textual support and stale-source risk rather than returning a single “faithful” judgment.

4 System

ContextTrace accepts a JSON trace containing query, answer, contexts, citations, and optional metadata. The implementation normalizes the trace, decomposes the answer into claims, discovers candidate evidence, evaluates support and contradiction, checks citation alignment, assesses source metadata, decides whether abstention was

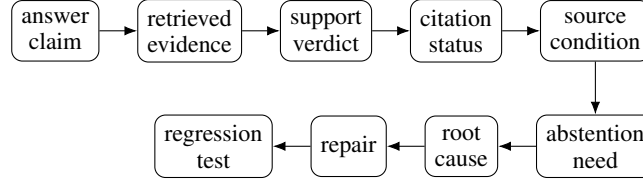


Figure 1: Evidence-chain forensics keeps grounding separate from source condition and downstream repair.

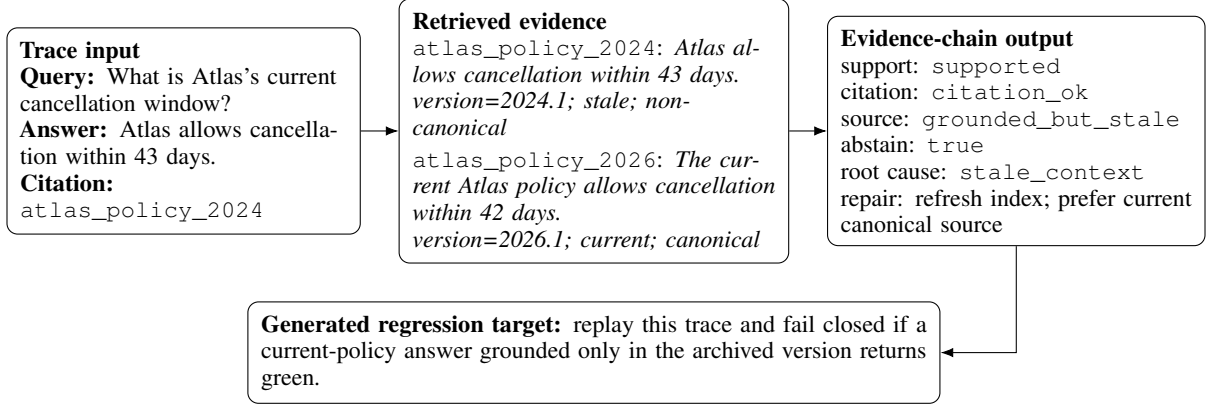


Figure 2: A groundedness-gap trace. The answer is verbatim-supported and correctly cited, but its source is stale for the current-policy query.

required, and infers a primary root cause. Each stage returns typed JSON consumed by the next stage and by the final report.

Claim verification combines lexical anchors with deterministic semantic normalization for entities, quantities, dates, negation, and common relational paraphrases. Contradiction checks are independent of citation checks: a claim may be contradicted by one source while citing a different source, or it may be supported by the retrieved set while citing the wrong source. Evidence-span localization records the source ID and minimal matched text. Source assessment uses freshness, authority, and canonical-source metadata only when provided. In the Atlas example (Figure 2), lexical support and citation checks remain positive; the explicit stale/current and canonical metadata is what changes the source state, triggers abstention, and selects `stale_context` as the root. This dependency is measured directly in Section 8 rather than treated as prose-level inference.

The diagnosis layer aggregates claim records into observable failure types and a primary root cause. Repair planning ties a suggested change to cited evidence and emits a regression-test target rather than claiming the change has already worked. Production profiles expose citation, contradiction, abstention, source, root-cause, and span modules as explicit switches; ablations therefore execute

the shipping verifier rather than a parallel research implementation.

The default pipeline is local and deterministic, with zero marginal model cost. Model-based judges are optional. A provider/model/prompt/cache/cost tuple must be frozen before such outputs enter a result table. We separately report a frozen GPT judge as an external baseline.

5 Benchmark Protocol

We preregister case sets, profiles, seeds, metrics, and claim gates in a machine-readable experiment specification. Prediction, scoring, review, and claim authorization are separate stages. Every bundle records case-ID hashes, candidate hashes, output hashes, dependency versions, runtime, and reported cost.

Datasets. Table 1 summarizes every split used in this paper; the paragraphs below give the role of each in full. Groundedness-Truth-Gap contains 180 author-curated, template-varied cases covering the distinctions exposed by the evidence-chain contract. It is designed to test whether a system exposes the distinctions required by source-aware RAG diagnostics. It is not intended to estimate natural-traffic accuracy. ContextTrace-Diag-150 contains 150 public-document traces covering sup-

port, contradiction, abstention, citation, source, and root-cause behavior. The naturalistic protocol has three disjoint roles. *Audit v1* is the original 50-case first-run boundary set and is used only for error taxonomy. The 40-case *dev* split covers ten source/citation/abstention failure families and is the only split used for rule development. The fresh *Eval v2* split contains 84 cases (12 categories, seven each); IDs, labels, source metadata, profile, scorer, and seed were hashed before its one scoring run. Naturalistic documents do not make authored queries, retrieval sets, and labels independent. The external primary track is a deterministic 200-case sample derived from RAGTruth (Niu et al., 2024). ARES and CRAG adapters provide secondary calibration tracks. The 500-case engineering suite supports module regression and ablation analysis; it is not independent validation.

Metrics. Primary diagnosis uses failure-label macro-F1 and dangerous false-green rate, where a false green is a trace with a material labeled failure that is reported as clean. Because Eval v2 contains 77 failures and only seven clean cases, we also report the full binary confusion counts, sensitivity, specificity, balanced accuracy, Matthews correlation coefficient (MCC), and Wilson intervals for sensitivity and specificity. These class-balanced statistics expose an always-failure baseline that otherwise obtains high binary F1. Diagnostic outputs use root-cause accuracy, citation-error F1, and token-level evidence-span overlap. Claim-verdict macro-F1 is reported separately because answer-level RAGTruth labels do not always determine an unambiguous claim-level projection. We report deterministic 95% case-bootstrap intervals with 400 draws for the frozen experiments. Audit v1 contains only failures; binary failure detection is therefore reported separately from label macro-F1 and is not treated as a balanced accuracy estimate. Eval v2 includes seven clean current-source cases and 77 failure-bearing cases; consequently, its specificity interval remains wide and the split does not substitute for a larger class-balanced evaluation.

The Atlas trace provides a running example of the annotation contract: support is supported, citation is correct, source condition is stale, abstention is true, and the root is stale_context. Eval v2’s seven grounded_but_stale cases instantiate the same field combination over public documents; they do not reuse the illustrative Atlas prose.

Review and claims. To make the evaluation auditable, we release a 50-case blinded annotation packet: 25 Eval v2 cases, 15 Groundedness-Truth-Gap cases, and 10 RAGTruth-derived cases. The packet uses opaque IDs and hides predictions, split membership, original case IDs, and private condition keys. The accompanying scorer reports agreement, kappa, adjudicated label changes, and subset-level metric deltas. Current labels are released with the packet and scorer so that independent review can reproduce, challenge, or revise them. A preregistered claim gate checks bundle integrity, external coverage, review status, thresholds, confidence intervals, same-ID competitor coverage, and Diag-150 sign-off before permitting broad cross-benchmark claims.

6 Experiments

The experiments ask whether evidence-chain diagnosis separates textual support from source-condition and abstention failures. We lead with the frozen naturalistic transfer result, then use external and stress controls for failure safety, the focused suite for contract conformance, and same-ID systems for comparison.

6.1 Naturalistic Boundary and Generalization

Table 3 distinguishes audit v1, dev, and fresh Eval v2. Audit v1 was never used for tuning. Dev-only rules improve source, root, citation, and abstention behavior without changing global support thresholds. Eval v2 was then frozen and scored once: ContextTrace catches 76 of 77 failures and preserves all seven clean cases (balanced accuracy 0.994, MCC 0.929, binary F1 0.993). Sensitivity is 0.987 (Wilson 95% CI [0.930, 0.998]); specificity is 1.000, but its interval is wide ([0.646, 1.000]) because there are only seven clean cases. Dangerous false-green is 0.012, source accuracy is 0.774, citation macro-F1 is 0.858, and abstention F1 is 0.932. Failure-label macro-F1 (0.457), root accuracy (0.583), and unverifiable F1 (0.000) miss target. The fresh naturalistic evaluation shows that failure detection transfers more reliably than fine-grained attribution: ContextTrace rarely marks unsafe answers clean, but source, root, and unverifiable labels remain imperfect. After inspection, a dev-backed rule for explicit replacement language removes the sole false green in a non-headline post-hoc audit (binary F1 1.000; false-green 0.000), while unverifiable F1 remains 0.000. We retain

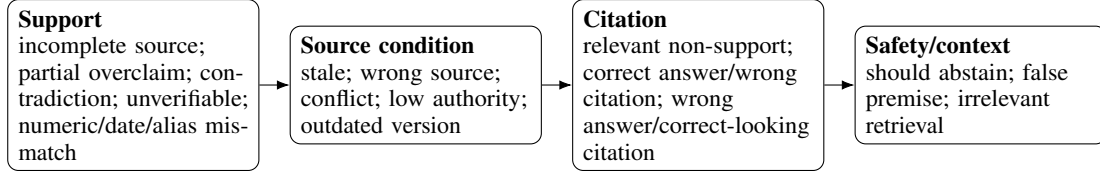


Figure 3: Groundedness-Truth-Gap taxonomy: 18 categories with 10 cases per category.

Split	n	Role	Status
Groundedness-Truth-Gap	180	Templated contract coverage	conformance
ContextTrace-Diag-150	150	Public-document diagnostic traces	control (sign-off pending)
Audit v1	50	First-run boundary audit	diagnostic (untuned)
Dev	40	Rule-development split	tuning
Eval v2	84	Frozen naturalistic evaluation	frozen-once
RAGTruth (full)	200	External calibration sample	control
RAGTruth (locked)	120	Locked control subset	control
Adversarial	90	Constructed stress set	control
Engineering suite	500	Regression suite	engineering (non-validation)

Table 1: Dataset splits and their roles. Only Dev is used for rule development; Audit v1 is retained for first-run error taxonomy, and the engineering suite is used only for regression testing.

the untouched first run as the primary naturalistic result.

The controlled-to-naturalistic gap is explicit in Tables 2 and 3: failure detection and abstention transfer more reliably than root attribution, while the narrow unverifiable label does not transfer. Clean specificity is preliminary, and the all-failure dev split cannot identify a useful scalar threshold.

6.2 Frozen RAGTruth and Stress Failure-Safety Controls

Frozen RAGTruth and the adversarial suite are failure-safety controls (Table 5). The targeted changes preserve full-set failure F1 at 0.955, false-green at 0.000, and root-cause accuracy at 0.955; the locked split remains 0.945/0.000/0.942. Claim F1 is secondary: RAGAS is higher than ContextTrace on the sparse locked claim-count subset (0.533 versus 0.367).

6.3 Groundedness-Truth-Gap as Controlled Conformance

Groundedness-Truth-Gap contains 180 frozen cases across 18 categories (Figure 3), with ten template-varied cases per category. Each records support, source, abstention, root, citation, evidence span, rationale, and a regression target. Groundedness-Truth-Gap acts as a taxonomy conformance suite: it verifies that the evidence-chain contract is implemented, not that the system generalizes to natural traffic.

Table 2 reports controlled diagnostic confor-

mance. Registered targets pass; the 0.944 source score reflects ten unavailable-fact cases labeled “unknown” whose metadata is current. We retain the frozen mismatch. Perfect failure and root scores are conformance results on templated cases.

6.4 Same-ID Baselines and GPT Judge Comparison

Table 4 reports base-rate-aware same-ID Eval v2 results. We freeze scalar thresholds on dev by minimizing false greens first and maximizing binary F1 second, then run Eval v2 once. Both lexical coverage and MiniLM cosine obtain 0.957 F1 and zero false greens, but predict failure for all 84 cases because dev contains no clean cases; their specificity and MCC are 0.000 and balanced accuracy is 0.500. Thus their high F1 is a base-rate artifact, not useful discrimination. We include a frozen GPT judge baseline for Eval v2. GPT-4.1-mini has the same confusion counts as ContextTrace (76 TP, 7 TN, 0 FP, 1 FN), with zero parse failures and an estimated evaluation cost of \$0.035. Its finer diagnoses are weaker: support macro-F1 is 0.430, root accuracy 0.333, citation macro-F1 0.346, abstention F1 0.364, and unverifiable F1 0.000. RAGAS 0.4.2 and DeepEval 4.0.5 installed successfully, but their faithfulness metrics required evaluator LLM credentials or a frozen local evaluator configuration; we therefore report unevaluated runs as N/A rather than treating them as failures. A local Phi-3 pilot was also infeasible: the first RAGTruth probe timed out after 92.8 seconds, so it was not promoted to a

frozen Eval v2 baseline. RAGAS and DeepEval are therefore not evidence for or against the system on Eval v2. On the separate locked RAGTruth track, cached same-ID RAGAS and generic-judge comparisons are reported as failure-safety controls, not as source-diagnostic baselines; Appendix Table 8 reports the available cross-system rows without comparing metrics across tracks. Beyond binary detection, ContextTrace produces local evidence-chain records and repair/regression targets.

6.5 LLM-Simulated Annotation Stress Test

We also run a frozen LLM-simulated annotation audit over the blinded 50-case packet. This is a model-based stress test, not ground truth. The model sees neither ContextTrace predictions nor author labels. Agreement with author labels is 0.900 for binary failure and 0.420 for both root and citation. Unverifiable agreement is nominally 0.900, but class imbalance makes this misleading: kappa is 0.000 and the model misses all five author-positive cases in the blinded 50-case packet’s unverifiable-labeled subset – a different set from the fourteen gold-unverifiable cases in the full 84-case Eval v2 split discussed in the error analysis. The latter count is two categories with seven cases each: `relevant_but_insufficient` and `citation_relevant_non_supporting`. Disagreements concentrate around missing versus not-applicable citations and source-conflict versus unsupported/abstention roots. We use these labels only to localize disagreement patterns, not as validation evidence.

7 Ablations

Nine registered RAGTruth development ablations test the shipping modules on a fixed 25-case sample. Removing contradiction checks has the largest observed cost and raises false greens; lexical-only scoring also degrades detection. Citation, source, and abstention removals are marginal on this small split, so we do not infer that those modules are generally irrelevant: a 25-case sample under-exercises citation, source, and abstention behavior, and Groundedness-Truth-Gap’s controlled conformance results (Table 2) show these same modules pass when case coverage is adequate. Full ablation rows are included in the supplementary artifact.

8 Error Analysis

The untouched verifier initially obtained 0.774 failure F1, 0.000 unverifiable F1, 0.762 abstention F1, and 0.833 root accuracy on the new benchmark. Seventy cases changed after four narrow rules: freshness-sensitive post-source abstention, stronger-source conflict precedence, explicit false-premise contradiction, and authoritative-current corpus-gap classification. No global support threshold or locked label changed. The main overfitting risk remains high because all affected cases share curated templates; unchanged RAGTruth gates provide only a regression control and do not establish generalization.

On the separate 90-case adversarial diagnostic set, a narrow verbal-antonym check raises failure F1 from 0.873 to 0.934 and reduces dangerous false-green from 0.067 to 0.000 while preserving the frozen RAGTruth gates. Six compound source-conflict cases still fail exact label-set matching: the system emits `should_have_abstained + stale_source`, whereas the frozen label is `contradicted_answer + stale_source`. Six wrong-answer cases expose both contradiction and citation mismatch where the frozen failure set contains only contradiction and abstention. These are visible diagnostic disagreements rather than false greens. Table 5 keeps both the improvement and residual error in view.

Naturalistic Eval v2’s untouched run contains one dangerous false green: an explicit “Settings instead of ServiceContext” replacement was mistaken for support because the legacy term appeared in the current source. A general, dev-backed replacement rule catches this case in a post-hoc audit; it is not reported as a fresh held-out result. This illustrates a boundary of deterministic lexical matching: replacement and deprecation language often mentions both obsolete and current entities. The narrow rule handles explicit “instead of,” “renamed,” and “replace X with Y” markers, but broader paraphrastic entailment remains outside the deterministic verifier. Most other misses are wrong fine-grained fields: incomplete authoritative contexts become partial support rather than unverifiable, and source precedence does not always select the labeled root. Misassigned root or source fields are lower risk than false-green outcomes when the trace is still blocked as unsafe, but they can still misdirect remediation.

Metric	Score	95% CI	<i>n</i>	Role
Failure macro-F1	1.000	[1.000, 1.000]	180	conformance
Support-verdict macro-F1	0.968	[0.949, 0.987]	180	descriptive
Source-condition accuracy	0.944	[0.911, 0.978]	180	conformance
Other registered fields (five)	1.000	[1.000, 1.000]	180	conformance
Dangerous false-green	0.000	[0.000, 0.000]	180	safety

Table 2: Controlled Groundedness-Truth-Gap conformance (18 categories, 10 cases each). “Other” summarizes unverifiable F1, abstention F1, root accuracy, citation F1, and regression coverage; this table tests implementation of an author-defined contract, not generalization.

Split	<i>n</i>	Label F1	Binary F1	Bal. acc.	MCC	False green	Source	Root	Citation	Unver.	Abstain	Regr.
Audit v1 (untuned first run)	50	0.325	1.000	N/A	N/A	0.000	0.240	0.340	0.250	0.000	0.730	1.000
Dev before (debug only)	40	0.433	1.000	N/A	N/A	0.000	0.550	0.475	0.572	0.000	0.737	1.000
Dev after (debug only)	40	0.574	1.000	N/A	N/A	0.000	0.850	0.675	0.639	0.000	0.909	1.000
Eval v2 (fresh, frozen)	84	0.457	0.993	0.994	0.929	0.012	0.774	0.583	0.858	0.000	0.932	1.000
Eval v2 (post-hoc audit)	84	0.466	1.000	1.000	1.000	0.000	0.774	0.595	0.858	0.000	0.923	1.000

Table 3: Naturalistic boundary. Development uses only dev; the frozen Eval v2 row remains primary over the labeled post-hoc audit. Scenarios and labels are author-created over public documents.

The frozen root confusion matrix in Table 6 makes this boundary concrete. Of 41 gold conflict-root cases, 23 are correct, 11 are assigned to stale context, and five to corpus gap. All seven corpus-gap cases and seven of the 14 wrong-citation cases collapse to the generic `should_have_abstained` root. The 0.583 root accuracy therefore hides systematic precedence errors rather than diffuse noise.

Naturalistic unverifiable performance. This is a taxonomy boundary, not a hidden clean-pass failure. All 14 gold-unverifiable cases have zero matched evidence terms. The conservative relevance rule therefore blocks them as unsupported/unsafe instead of assigning the narrower unverifiable label. Fail-closed behavior is preserved, but the fine-grained attribution target fails. We exclude reliable naturalistic unverifiable attribution from our claims and make it the primary target for a future hybrid entailment/gap-detection profile. Such a profile is not implemented here.

Source metadata and temporal-cue counterfactuals. Table 7 reports post-review diagnostics, not new held-out results. Replaying the current verifier after removing source-group, freshness, version, timestamp, authority, and canonicity metadata drops source-condition accuracy from 0.774 to 0.000; every source condition becomes `unknown`. Binary balanced accuracy remains 0.974 because other support/citation checks still catch 73 of 77 failures. In a paired ablation of the seven stale cases, removing temporal cue words from the query alone leaves only three correct stale-source diag-

noses and two correct abstention decisions. A stronger removal from query, answer, and citation claim still yields only three correct decisions for each field. This quantifies that source diagnosis does not currently infer provenance relations robustly from prose alone.

External CRAG case study. We further examined source-condition detection in a 14-case CRAG study (Yang et al., 2024) by injecting staleness, version, and authority conflicts into retrieved contexts and deriving labels from the injection procedure. Across 200 CRAG rows, manual scrutiny identified only one clean same-quantity authority conflict and two organic staleness pairs. The study also exposed two structural dependencies: recognized machine-readable metadata yields only three source hits in five cases and one correct abstention; unrecognized metadata values yield zero of three, while organic or authored prose-only signals yield zero of three. Three clean controls abstain correctly but their source condition is `unknown`, not `current`. This qualitative evidence identifies conflict scarcity and vocabulary-bound metadata handling; it does not estimate naturalistic generalization.

9 Discussion

The strongest result is operational: ContextTrace catches 76 of 77 failure-bearing Eval v2 cases without rejecting any of the seven clean cases, while producing deterministic local records with root, citation, abstention, repair, and regression-test structure. A frozen GPT-4.1-mini judge happens to pro-

System	TP	TN	FP	FN	F1	Bal. acc.	Specificity	MCC	Diagnostic output
ContextTrace (frozen)	76	7	0	1	0.993	0.994	1.000	0.929	full chain
GPT-4.1-mini (frozen)	76	7	0	1	0.993	0.994	1.000	0.929	judge fields
Lexical coverage	77	0	7	0	0.957	0.500	0.000	0.000	scalar only
MiniLM cosine	77	0	7	0	0.957	0.500	0.000	0.000	scalar only
RAGAS 0.4.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	not run
DeepEval 4.0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	not run

Table 4: Same-ID Eval v2 binary results. Failure prevalence is 77/84. ContextTrace/GPT specificity has Wilson 95% CI [0.646, 1.000]; the scalar baselines predict failure for every case. N/A means not run, not zero.

Track / profile	n	Failure F1	False green	Root	Role
RAGTruth full / diagnostic-safe	200	0.955	0.000	0.955	external control
RAGTruth locked / diagnostic-safe	120	0.945	0.000	0.942	locked control
Adversarial stress / before	90	0.873	0.067	0.933	error discovery
Adversarial stress / after	90	0.934	0.000	1.000	targeted regression

Table 5: Failure-safety controls. RAGTruth passes frozen gates; the constructed stress suite records the targeted fix and residual exact-label disagreements.

duce the same binary confusion counts; this is parity on one small, failure-heavy split, not evidence of evaluator equivalence. Groundedness-Truth-Gap checks that contract under controlled cases; Eval v2 exposes its transfer boundary.

The split naturalistic protocol changes the strength of that conclusion. Eval v2 largely remains conservative at trace level and improves source and citation assignment over audit v1, but label, root, and unverifiable behavior still miss preregistered targets. The system is therefore best characterized as a reproducible diagnostic framework whose failure blocking transfers more reliably than its fine-grained attribution. The metadata and temporal-cue counterfactuals sharpen that boundary: source-condition diagnosis is currently a verifier of supplied provenance signals, not a robust prose-level provenance inference system.

The RAGAS comparison is deliberately non-adversarial. RAGAS remains stronger on the sparse RAGTruth claim-count subset, while ContextTrace is designed for failure safety and inspectable diagnostics. The claim-calibrated profile gains only 0.020 full-set claim F1 and drops failure F1 to 0.699, so safety gates exclude it from headline results.

Operational contract. Unlike scalar baselines, ContextTrace emits structured source, citation, root, abstention, repair, and regression-test records. Unlike the GPT judge baseline, it runs locally, supports deterministic replay, and requires no additional model calls. The evaluation also includes production-path ablations. This comparison is a

systems tradeoff rather than a general claim against model judges.

Claim boundary. We limit our claims to dataset-specific diagnostics and reproducible evidence-chain behavior. We do not claim broad cross-benchmark superiority, independent validation, improved human actionability, or real-world truth verification. The blinded packet is a step toward independent adjudication rather than a substitute for it.

Reproducibility. The workflow pins IDs, profiles, bootstrap seed, candidate files, hashes, runtime, and review status. The packet, scorer, current labels, and hashes accompany the paper as supplementary material.

10 Conclusion

ContextTrace represents RAG failures as inspectable chains connecting answer claims, evidence spans, citations, source conditions, abstention, root causes, and repairs. Its deterministic pipeline and fail-closed evidence gates produce auditable evidence-chain records, but source-condition attribution currently depends strongly on explicit metadata and temporal cues. Groundedness-Truth-Gap provides controlled conformance coverage, frozen RAGTruth provides a failure-safety control, and Eval v2 exposes weaker fine-grained naturalistic attribution. Together, these findings position the contribution as reproducible diagnosis rather than truth verification. Independent adjudication remains necessary.

Limitations

This paper evaluates ContextTrace as a reproducible diagnostic framework rather than as a completed independent human validation study. Independent adjudication of the RAGTruth-derived and ContextTrace-Diag-150 labels is still in progress, and the planned human actionability study remains future work. We therefore interpret the reported metrics as author-labeled calibration results, while releasing the blinded packet and scorer to support external review.

Groundedness-Truth-Gap, ContextTrace-Diag-150, and the adversarial suite are synthetic or author-curated, and are designed to test coverage of the proposed evidence-chain contract. Template variation helps exercise the taxonomy, but it does not by itself measure performance on naturally occurring traffic. The public-document Eval v2 split provides a transfer check beyond the controlled suite, but its scenarios and labels are still author-created and cannot estimate real-world failure prevalence. Its 77/7 failure/clean imbalance makes binary F1 optimistic for degenerate always-failure systems; balanced accuracy and MCC correct this comparison, but specificity is still estimated from only seven clean cases (Wilson 95% CI [0.646, 1.000]). Fine-grained naturalistic attribution remains weaker than binary failure detection, especially for unverifiable cases. The post-hoc replacement rule addresses one observed false green but does not alter the primary frozen Eval v2 result.

The external controls also have bounded scope. The RAGTruth experiment uses a deterministic 200-case sample rather than the full corpus; ARES/NQ is used as an upstream-label calibration; and the current CRAG adapter uses gold answers as a grounding proxy pending source-grounding review. Answer-level labels do not always define atomic claim boundaries, and only a small number of cases have explicit claim-count labels, making claim F1 unsuitable as a headline metric. The evidence-span metric uses token overlap and does not directly measure span minimality or multi-span evidence recovery.

The verifier is deterministic and interpretable, but it is not a general entailment model. It handles lexical anchors, normalized entities, quantities, dates, negation, and common paraphrases, but can miss implicit reasoning, paraphrastic deprecation, long dependency chains, domain-specific numerical relations, multilingual traces, and evi-

dence distributed across many documents. Source trust, freshness, and incompleteness also depend on metadata that may be missing or inconsistent in production traces. In our post-review counterfactual, stripping such metadata reduces source accuracy from 0.774 to 0.000; removing temporal cue words from the queries of seven stale cases leaves only three correct source decisions and two correct abstention decisions. These diagnostics were designed after review and are limitation probes, not held-out evidence. A supported claim should therefore be read as supported by the supplied evidence, not as verified truth in the world.

Several comparison and ablation gaps remain. NLI-only and judge-only module ablations were not registered as frozen replacements, same-ID baseline APIs expose fewer diagnostic fields, and the frozen GPT judge is a model baseline rather than an independent annotator. MiniLM cosine is included, but the all-failure dev split makes its specificity threshold non-identifiable. Cost and latency estimates are environment-dependent. The supplementary artifact provides the audit packet, scorer, and recorded result files, but cannot redistribute private condition keys, human identities, or future licensed response data.

Ethical Considerations

ContextTrace can process traces locally, reducing the need to transmit prompts, answers, or retrieved documents to an external evaluator. Local execution does not remove obligations around access control, retention, dataset licenses, or personal information contained in traces. Users should minimize stored content, restrict report access, and avoid publishing sensitive evidence spans.

Automatic diagnoses can be wrong and should not be the sole basis for decisions in medical, legal, financial, employment, or other high-impact settings. A “supported” result means supported by supplied evidence, not true in the world. Repair suggestions require engineering validation and a recaptured passing trace.

The simulated-review pilot is explicitly labeled to prevent model outputs from being misrepresented as human judgments. Future human participation must be voluntary, appropriately reviewed where required, compensated consistently, and reported without identifying reviewers in the paper. Raw responses and disagreements should be retained rather than selectively removed.

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References

- Arize AI. 2026. [OpenInference: Semantic conventions for AI observability](#). Open-source specification.
- Sizhe Cheng, Jiaping Li, Huanchen Wang, and Yuxin Ma. 2025. [RAGTrace: Understanding and refining retrieval-generation dynamics in retrieval-augmented generation](#). *Preprint*, arXiv:2508.06056.
- Cloud Native Computing Foundation. 2026. [OpenTelemetry specification](#). Technical specification.
- Dvir Cohen, Tamir Houri, Lin Burg, and Gilad Barkan. 2025. [RAGXplain: From explainable evaluation to actionable guidance of RAG pipelines](#). *Preprint*, arXiv:2505.13538.
- Confident AI. 2026. [DeepEval: The evaluation framework for large language models](#). Software documentation.
- Darshan Deshpande, Varun Gangal, Hersh Mehta, Jitin Krishnan, Anand Kannappan, and Rebecca Qian. 2025. [TRAIL: Trace reasoning and agentic issue localization](#). *Preprint*, arXiv:2505.08638.
- Shahul Es, Jithin James, Luis Espinosa Anke, and Steven Schockaert. 2024. [RAGAS: Automated evaluation of retrieval augmented generation](#). In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–158.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. [Retrieval-augmented generation for knowledge-intensive NLP tasks](#). In *Advances in Neural Information Processing Systems 33*.
- Kenichirou Narita, Siqi Peng, Taku Fukui, Moyuru Yamada, Satoshi Munakata, and Satoru Takahashi. 2026. [Overcoming the “impracticality” of RAG: Proposing a real-world benchmark and multi-dimensional diagnostic framework](#). *Preprint*, arXiv:2604.02640.
- Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. 2024. [RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 10862–10878.
- Dongyu Ru, Lin Qiu, Xiangkun Hu, Tianhang Zhang, Peng Shi, Shuaichen Chang, Jiayang Cheng, Cunxiang Wang, Shichao Sun, Huanyu Li, Zizhao Zhang, Binjie Wang, Jiarong Jiang, Tong He, Zhiguo Wang, Pengfei Liu, Yue Zhang, and Zheng Zhang. 2024. [RAGChecker: A fine-grained framework for diagnosing retrieval-augmented generation](#). In *Advances in Neural Information Processing Systems 37*, pages 21997–22027.
- Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. 2024. [ARES: An automated evaluation framework for retrieval-augmented generation systems](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 338–354.
- Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Lingpeng Kong, Qi Liu, Tianyu Liu, and Zhifang Sui. 2024. [Large language models are not fair evaluators](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 9440–9450.
- Xiao Yang, Kai Sun, Hao Xin, Yushi Sun, Nikita Bhalla, Xiangsen Chen, Sajal Choudhary, Rongze Daniel Gui, Ziran Will Jiang, Ziyu Jiang, Lingkun Kong, Brian Moran, Jiaqi Wang, Yifan Ethan Xu, An Yan, Chenyu Yang, Eting Yuan, Hanwen Zha, Nan Tang, and 8 others. 2024. [CRAG – comprehensive RAG benchmark](#). In *Advances in Neural Information Processing Systems 37*, pages 10470–10490.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, Hao Zhang, Joseph Gonzalez, and Ion Stoica. 2023. [Judging LLM-as-a-judge with MT-Bench and chatbot arena](#). In *Advances in Neural Information Processing Systems 36*, pages 46595–46623.

A Post-Review Diagnostic Detail

Gold \ predicted	Answer overreach	Conflict	Corpus gap	No failure	Should abstain	Stale	Wrong citation
Answer overreach	7	0	0	0	0	0	0
Conflict	0	23	5	1	0	11	1
Corpus gap	0	0	0	0	7	0	0
No failure	0	0	0	7	0	0	0
Should abstain	0	0	0	0	1	0	0
Stale	0	1	1	0	0	4	1
Wrong citation	0	0	0	0	7	0	7

Table 6: Frozen Eval v2 root-cause confusion matrix (84 cases). Rows are gold labels and columns are predictions; diagonal counts are correct.

Diagnostic arm	<i>n</i>	Source correct	Abstention correct	Binary bal. acc.	Status
Eval v2, full metadata (current verifier)	84	65/84	75/84	1.000	post-review replay
Eval v2, source metadata removed	84	0/84	68/84	0.974	counterfactual
Eval v2 stale cases, query cues removed	7	3/7	2/7	N/A	counterfactual
Eval v2 stale cases, all cues removed	7	3/7	3/7	N/A	counterfactual
CRAG, recognized machine metadata	5	3/5	1/5	N/A	frozen case study
CRAG, unrecognized metadata values	3	0/3	0/3	N/A	frozen case study
CRAG, prose-only staleness	3	0/3	0/3	N/A	frozen case study
CRAG, clean no-metadata controls	3	0/3	3/3	N/A	frozen case study

Table 7: Source-condition dependence diagnostics. Counterfactuals were designed after review and are not held-out results; the CRAG arms contain only 3–5 cases each.

Track	System	<i>n</i>	Failure F1 [95% CI]	Claim F1	False green	Root	p95 ms	Cost/100
RAGTruth locked test	ContextTrace diagnostic_safe	120	0.945 [0.877, 0.983]	0.367	0.000	0.942	4293.477	0.000
RAGTruth locked test	RAGAS	120	0.150 [0.138, 0.202]	0.533	0.308	N/A	50060.891	N/A
RAGTruth locked test	Generic LLM judge	120	0.155 [0.129, 0.253]	0.400	0.275	0.525	5123.763	0.113
ARES/NQ 200	RAGAS	200	0.471 [0.426, 0.513]	N/A	0.030	N/A	14821.814	N/A
ARES/NQ 200	DeepEval	200	0.388 [0.342, 0.431]	N/A	0.360	N/A	7934.588	N/A
CRAG 200 proxy	RAGChecker	200	0.212 [0.185, 0.232]	N/A	0.000	N/A	N/A	N/A

Table 8: Same-ID baseline comparison. Systems are compared only within track; unsupported fields are N/A.